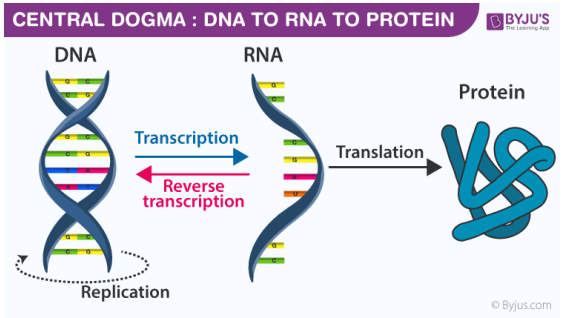
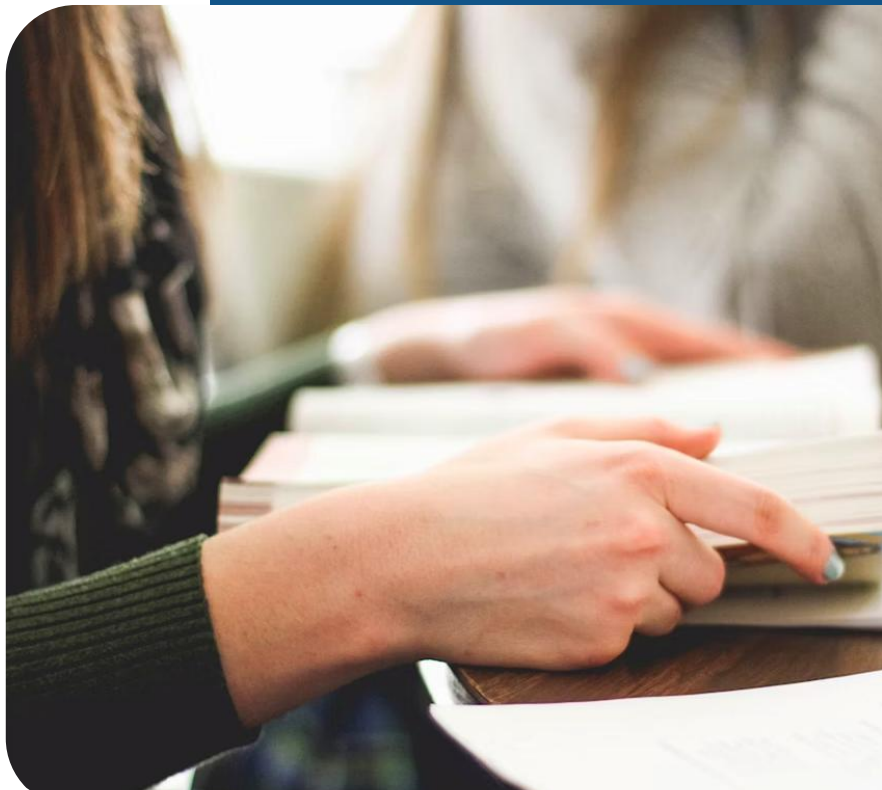
Central Dogma of Biology

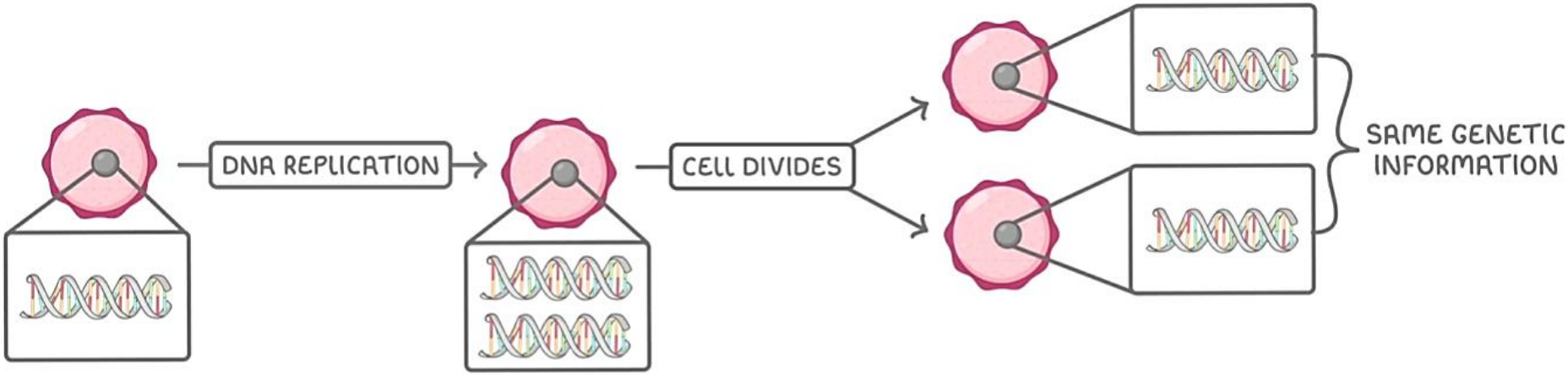


Replication

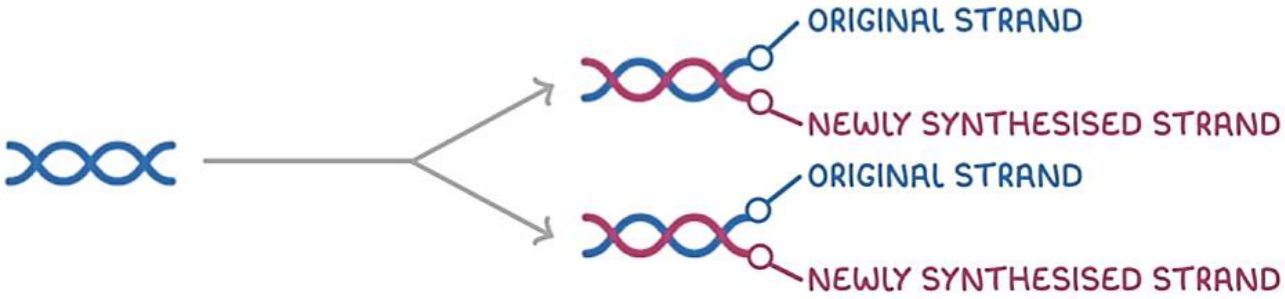


WHAT DNA REPLICATION IS

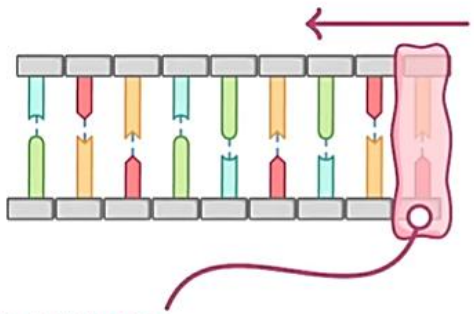
DNA MAKES A COPY OF ITSELF BEFORE CELL DIVISION



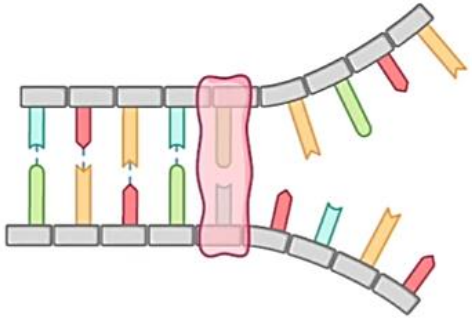
SEMI-CONSERVATIVE DNA REPLICATION



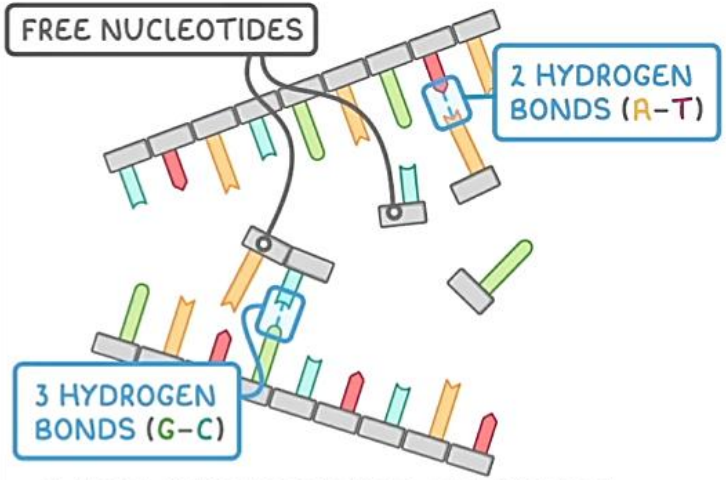
HOW DNA REPLICATION WORKS



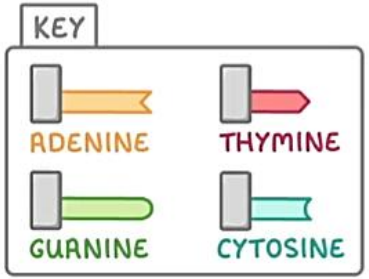
1. DNA HELICASE BREAKS HYDROGEN BONDS BETWEEN COMPLEMENTARY BASE PAIRS

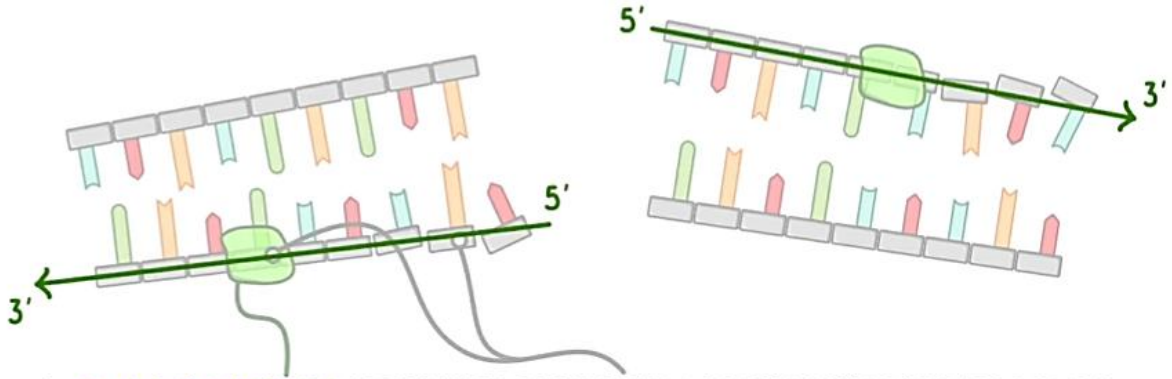


2. DOUBLE HELIX UNWINDS (2 DNA STRANDS SEPARATE)



3. EACH STRAND ACTS AS A TEMPLATE FOR FREE NUCLEOTIDES TO BIND TO

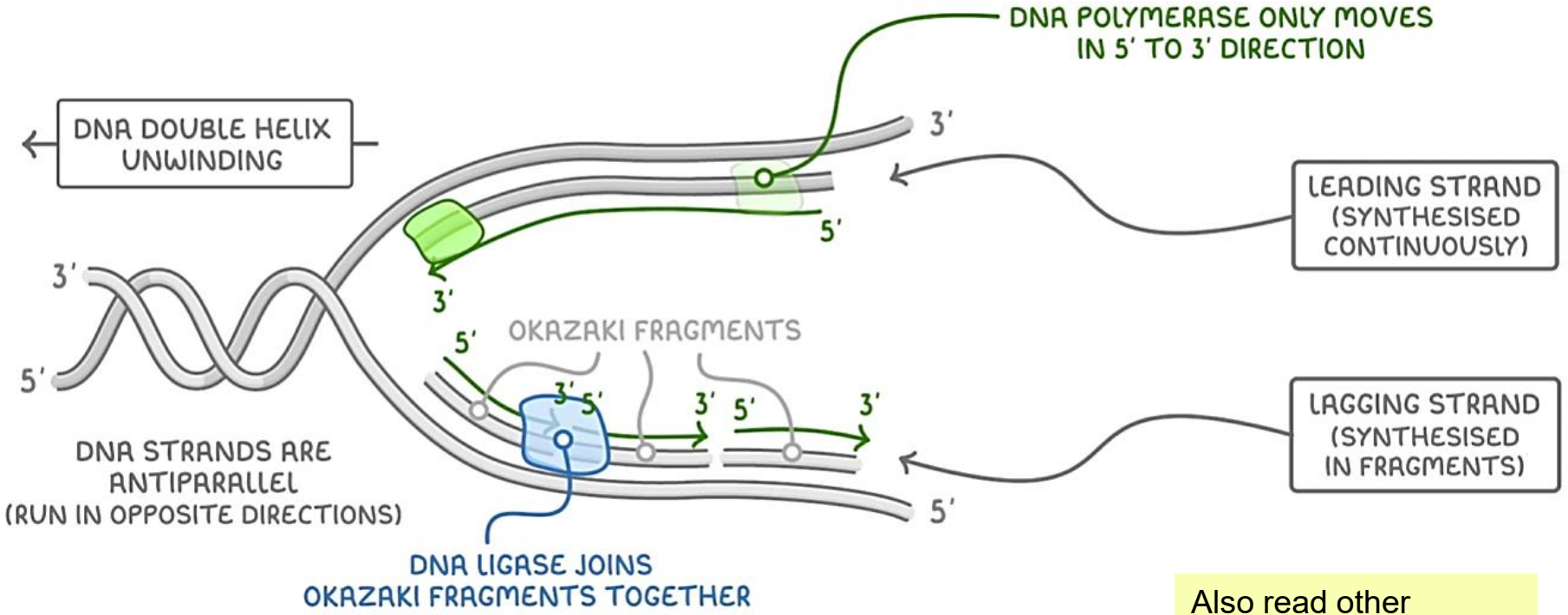




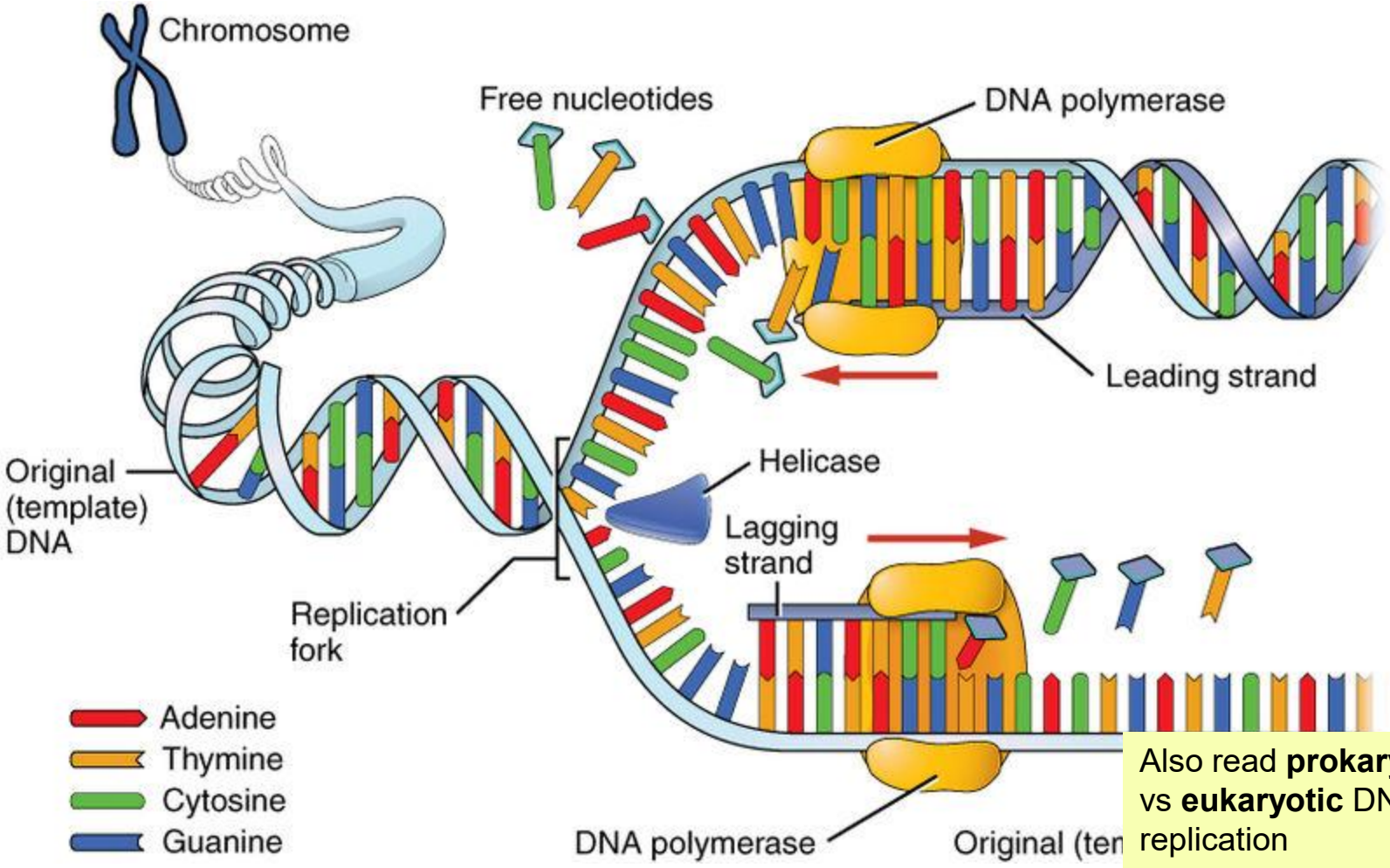
4. DNA POLYMERASE CATALYSES FORMATION OF PHOSPHODIESTER BONDS (ALWAYS MOVES 5' TO 3') (CREATES SUGAR-PHOSPHATE BACKBONES)

HOW LEADING & LAGGING STRANDS ARE SYNTHESISED

NEW DNA STRANDS SYNTHESISED DURING DNA REPLICATION



Also read other enzymes that involve in DNA replication



Also read **prokaryotic** vs **eukaryotic** DNA replication